

CORONARY AND PERIPHERAL ARTERY HEMODYNAMICS

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Selective coronary angiography is one of the most commonly performed procedures in the United States. Cardiologists who perform coronary angiography must be capable of promptly recognizing abnormalities in pressure waveforms sampled from the catheter tip during selective coronary angiography. This is crucial for patient safety, proper diagnosis, and appropriate intervention. In addition, it is important for cardiologists to understand the more sophisticated coronary hemodynamic assessments that allow an invasive assessment of coronary physiology. These techniques serve as important adjuncts to angiography and have become indispensable to the routine practice of interventional cardiology.

CORONARY CATHETER PRESSURE WAVEFORMS

Analysis of the pressure waveforms generated from the tip of the catheter during the performance of selective coronary angiography can provide valuable information to the operator. On engagement of a catheter into the ostium of either the right or left coronary artery, the physician should immediately review the catheter-tip pressure tracing before proceeding with contrast injection. This tracing normally appears as a typical aortic pressure waveform. An alert angiographer seeks two abnormalities in the pressure waveform: damping and ventricularization. Recognizing these abnormalities is essential, enhances the information provided by the angiogram, and may help avoid potentially serious complications of selective angiography.

Pressure *damping* refers to a drop in both the systolic and diastolic pressures with a loss of the usual features of an arterial pressure waveform. Damping of the waveform often indicates a narrowing of the coronary ostium. An example of this phenomenon is shown in Fig. 14.1. In this case the catheter pressure showed marked damping on the engagement of the left coronary artery (Fig. 14.1A). The angiogram showed modest narrowing of the ostium of the left main stem (Fig. 14.1B). Removal of the catheter from the left main stem reestablished the normal aortic pressure (Fig. 14.1C).

Ventricularization of the catheter-tip pressure is seen when the diastolic pressure drops on the engagement of the catheter and shifts appearance from the usual arterial waveform to one with a ventricular waveform morphology (Fig. 14.2). Pure ventricularization, in which there is only a drop in diastolic pressure and not in systolic pressure, implies a precise match between the diameters of the artery and the catheter lumen that leads to a coronary artery “wedge” pressure, reflecting left ventricular pressure (Fig. 14.3). More typically, the pressure trace appears both ventricularized and damped (see Fig. 14.2). Both ventricularization and/or damping may be obvious; sometimes the change is subtle (Fig. 14.4).

Both damping and ventricularization have similar causes (Box 14.1); both imply that the catheter tip plugs and obstructs the coronary artery lumen. The most common causes are the presence of obstructive disease at the coronary ostium and selective engagement of a smaller artery, such as the conus branch of the right coronary artery. On observing these abnormal waveforms, the angiographer should determine the cause before proceeding with coronary arteriography in the usual manner. Ventricularization or damping of the catheter pressure waveform may be the only clue to the existence of a significant ostial stenosis not appreciated by angiography. More importantly, the injection of contrast into a catheter with a damped or ventricularized pressure tracing might result in serious complications. For instance, selective and forceful injection of a full syringe of contrast into a conus branch of the right coronary artery or a small nondominant artery may lead to ventricular fibrillation or asystole. Similarly, if damping or ventricularization is caused by the presence of a small caliber coronary artery, then the naive angiographer might inject excessive contrast, causing an arrhythmia. Furthermore, if significant atherosclerotic disease is present at the ostium, a carelessly performed contrast injection into a damped catheter could cause hydraulic dissection of the proximal artery and result in abrupt closure of the vessel, with potentially serious or even fatal consequences. In the event that damping is caused by the presence of air or thrombus in the catheter, an operator unaware of this finding might inadvertently inject these into the artery causing distal embolization and ischemic complications. Thus the angiographer should be constantly vigilant for these abnormal pressure waveforms.

When damping or ventricularization of the catheter pressure waveform is observed, the catheter may first be withdrawn into the aortic root and contrast injected into the aortic cusp to determine the presence of ostial disease. Alternatively, a very small amount of contrast may be gently injected into the damped coronary catheter to determine the cause. If coronary ostial spasm is suspected, administration of intraarterial or sublingual nitroglycerin can lead to improvement in the spasm and normalize the waveform and hemodynamics. Should an air or thrombus be suspected in the catheter system, the system should be aspirated and thoroughly cleared before the catheter is reengaged and an injection is performed.

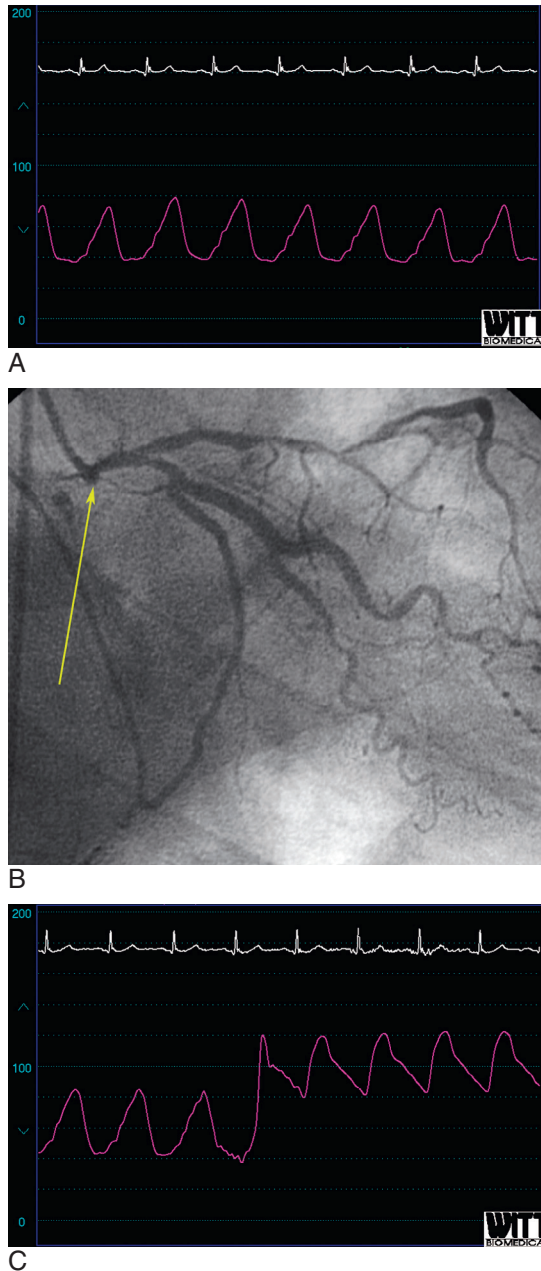


Fig. 14.1 (A) Example of pressure damping and ventricularization observed following engagement of the left coronary catheter. (B) The angiogram showed at least moderate disease of the ostium of the left main coronary artery (*arrow*) as well as significant disease in the left anterior descending artery. (C) When the catheter was withdrawn into the aorta, the pressure waveform was restored.



Fig. 14.2 Example of ventricularization of the coronary catheter pressure waveform.

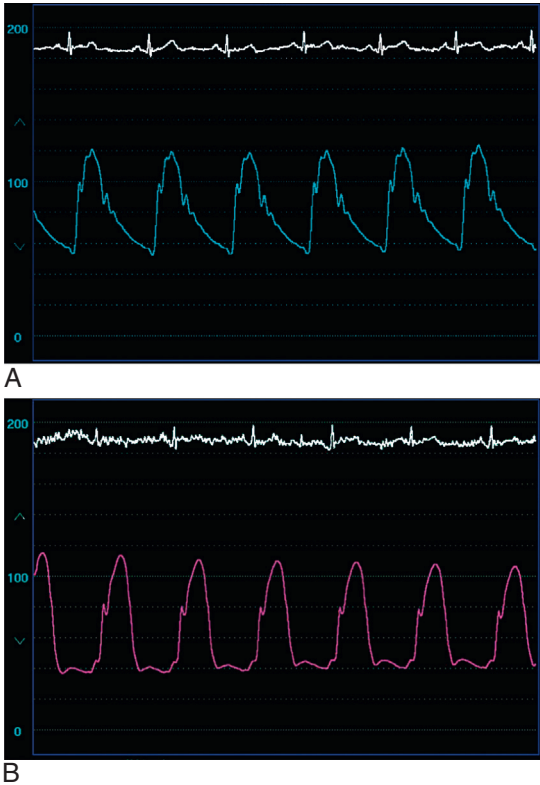


Fig. 14.3 Example of pure ventricularization observed after the engagement of a coronary catheter. (A) The normal-appearing aortic pressure waveform. (B) Changes to a ventricular waveform with a little drop in systolic pressure.

CORONARY HEMODYNAMICS

Although coronary angiography forms the basis for revascularization decisions in patients with coronary artery disease, there are limitations to this method that may impair its utility in the diagnosis of ischemic syndromes. Coronary angiography provides purely anatomic information and high-resolution, two-dimensional images of the arterial lumen but does



Fig. 14.4 An example of subtle ventricularization. (A) The catheter pressure demonstrated a normal aortic waveform prior to engagement. (B) On engagement it showed a loss of the dicrotic notch, narrowing of the waveform, and a slight drop in diastolic pressure consistent with mild ventricularization.

not provide any functional or physiologic information regarding the hemodynamic impact of a coronary lesion seen on angiography. Furthermore, angiography is influenced by injection technique, the patient's body habitus, the presence of overlapping vessel segments, and arterial tortuosity. Coronary angiography is most helpful when the angiogram reveals either normal arteries or the presence of a severely stenotic lesion. However, it is not uncommon that a clinician is left with questions regarding the significance of findings seen on angiography. Particularly troublesome scenarios include eccentric lesions, diffuse disease, or lesions of intermediate stenosis severity (40%–70% narrowing). Sophisticated coronary hemodynamic measurements are possible and allow precise flow and pressure assessments in the coronary arteries. Adjunctive diagnostic testing provided by coronary hemodynamic assessment helps determine the physiologic significance of ambiguous lesions seen on angiography and can guide revascularization decisions.

Box 14.1 Common Causes of Damping and/or Ventricularization of Catheter Pressure Waveform During Coronary Angiography

During the Engagement of the Right Coronary Artery

1. Atherosclerotic disease of the ostium of the RCA
2. Catheter-induced spasm of the coronary artery
3. Selective engagement of the conus branch of the RCA
4. Total occlusion of the RCA
5. Engagement of a small, nondominant artery

During the Engagement of the Left Coronary Artery

Atherosclerotic disease of the ostium of the left main stem
Deep seating of the catheter and selective engagement of either the LAD or LCX

During the Engagement of Either Coronary Artery

Presence of a small caliber artery with match in size between the diameter of the catheter and the coronary artery
Kinking of the catheter during catheter manipulation
Malposition of the catheter against the wall of the aorta
Presence of thrombus or air bubble in the catheter

LAD, Left anterior descending; LCX, left circumflex; RCA, right coronary artery.

CORONARY PHYSIOLOGY

An increase in myocardial oxygen demand, caused by an increase in wall tension, contractility, or heart rate, requires an increase in oxygen supply or else results in myocardial ischemia. Since myocardial oxygen extraction from blood is already near maximal, the only significant mechanism to satisfy an increase in oxygen demand is to increase coronary blood flow to the myocardium. Coronary blood flow is regulated by changes in coronary vascular resistance. The epicardial coronary arteries function primarily as conductance vessels, accounting for only 5% of total coronary vascular resistance; the remaining 95% of resistance across the coronary bed is provided by the small (<300 μm) intramyocardial arterioles, and thus these are the major regulators of coronary blood flow.

These small resistance vessels are influenced by multiple regulators, including endothelium-derived agents (nitrous oxide, endothelin-dependent hyperpolarizing factor, prostacyclin, and endothelin), metabolites (adenosine, hypoxia, hypercapnia, and acidosis) and neurohormonal mechanisms. The process of autoregulation provides fine control of the resistance vessels, maintaining myocardial blood flow over the relatively wide range of coronary perfusion pressures seen with physiologic changes in blood pressure. Thus in normal individuals myocardial perfusion is maintained when the mean aortic pressure is between 50 and 150 mm Hg. Outside of this range, however, autoregulatory mechanisms fail, and coronary blood flow depends on perfusion pressure (Fig. 14.5).

EFFECTS OF CORONARY STENOSIS ON CORONARY FLOW

Resting blood flow in the left and right coronary arteries under normal conditions averages 0.5 to 1.0 mL/min/g and 0.3 to 0.6 mL/min/g, respectively. In the absence of a stenosis, coronary blood flow can increase four to five times above resting flow under conditions of high myocardial oxygen demand (Fig. 14.6). This ability to increase blood flow over the resting state is termed *coronary flow reserve*. In the presence of a stenosis, a pressure drop will occur in a coronary artery; however, autoregulation will attempt to preserve blood flow by reducing the resistance of the small arterioles. Resting myocardial blood flow can be maintained over a wide range of stenosis. Eventually, when stenosis is severe, autoregulatory mechanisms become exhausted, and myocardial blood flow decreases. During conditions of increased myocardial oxygen demand, however, autoregulatory mechanisms fail at much lower stenosis severity. Accordingly, coronary flow reserve diminishes at less severe degrees of stenosis than resting coronary blood flow. Therefore a deficiency of coronary flow reserve implies maximal vasodilation of the resistance vessels and the presence of hemodynamically significant stenosis.

CORONARY FLOW RESERVE

Several different terms describe different aspects of coronary flow reserve. *Absolute flow reserve* describes the ratio of hyperemic flow in a stenotic artery to resting flow in the same artery. This is the basis of the Doppler-derived method for invasively determining coronary flow reserve. *Relative flow reserve* is the term used to describe the ratio of hyperemic flow in a stenotic artery to hyperemic flow in a normal artery. This is the primary principle on which most noninvasive nuclear perfusion imaging techniques are based. Importantly, relative flow reserve depends on at least one normal vascular territory; therefore these techniques may be limited in the presence of multivessel coronary disease. Finally, *fractional flow reserve* (FFR) is the term used to describe the ratio of the maximum achievable flow in the presence of a stenosis to the theoretical maximum flow in the same vessel in the absence of a stenosis.

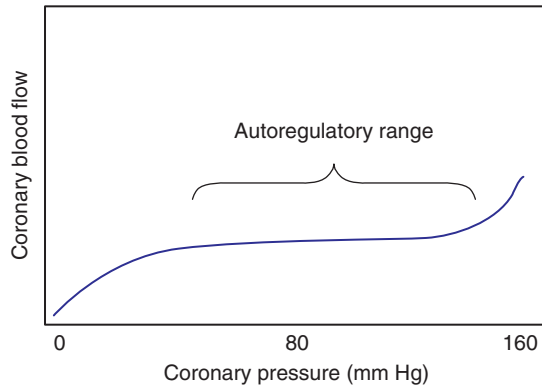


Fig. 14.5 Autoregulation of coronary blood flow. Schematic representation of the phenomenon of autoregulation, which states that coronary blood flow is maintained over the physiologic range of coronary pressure.

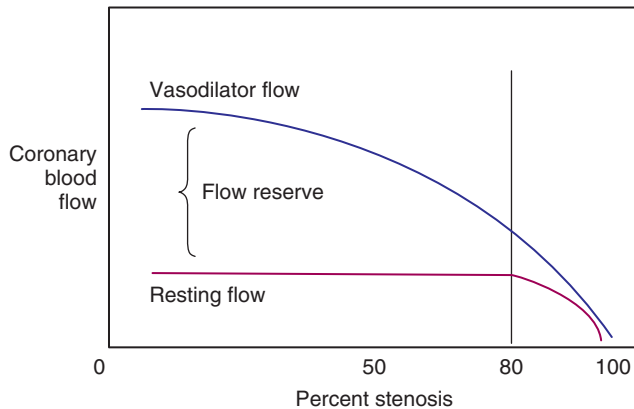
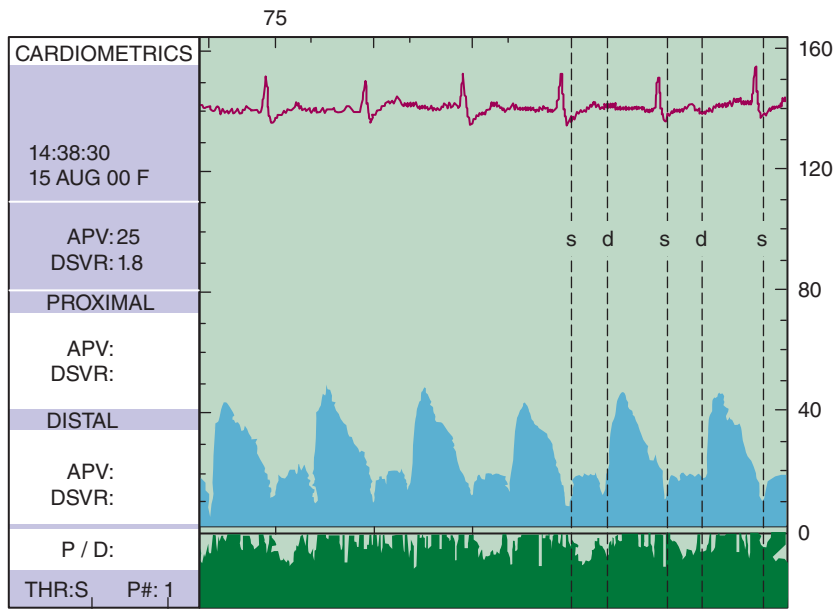


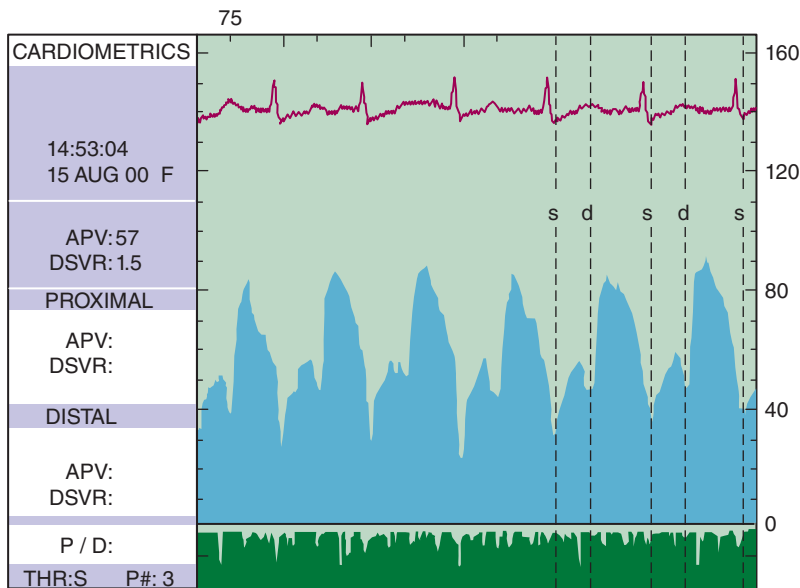
Fig. 14.6 Relation of stenosis severity to coronary blood flow. At rest coronary blood flow is maintained despite progressive coronary stenosis, until severe (>80%) stenosis occurs. During vasodilator stress, blood flow decreases from maximum flow for a constant vessel severity. The ability to augment flow from resting flow rates is known as *flow reserve*.

DOPPLER-DERIVED ABSOLUTE FLOW RESERVE

A 0.014-inch angioplasty guidewire outfitted with an ultrasound crystal at the tip, capable of generating a Doppler signal and measuring blood flow velocity, provides the means to estimate absolute flow reserve in the cardiac catheterization laboratory (Fig. 14.7).¹⁻³ This method assumes that blood velocity is proportional to blood flow for a constant vessel area. The technique involves positioning the tip of the Doppler wire past the stenosis and obtaining a satisfactory velocity envelope. The wire tip samples blood velocity a few millimeters away from the tip, typically expressed as the average peak velocity (APV). Normal resting APV is 15 to 30 cm/s. Without moving the wire tip, hyperemia is induced either from intravenous administration of adenosine (140 µg/kg per min for at least 2–4 min) or from intracoronary administration of bolus adenosine (20–100 µg bolus). The APV measurement is repeated during peak hyperemia. Coronary flow reserve (or more accurately coronary velocity reserve) is described as the ratio of hyperemic APV to resting APV. Normal coronary velocity reserve is at least 2.0, and values up to 5.0 may be observed. In the setting of moderate stenosis, coronary flow reserve <2.0 by the Doppler technique correlates with ischemia by nuclear perfusion techniques.¹⁻³ The Doppler method is rarely used in the current era for clinical purposes, mainly because of several important limitations. This methodology is dependent on obtaining a stable, high-quality Doppler signal, which is not always feasible, particularly in the presence of vessel tortuosity or large branches. Intermittent loss of signal because of tenuous wire position against the wall of the artery may frustrate the operator. More importantly, however, the APV varies with blood pressure and heart rate and changes in these variables during the measurements influence the results.⁴ In addition, conditions



A



B

Fig. 14.7 Example of the Doppler method of measuring coronary flow reserve. (A) The average peak velocity at baseline was 25 cm/s. (B) Intravenous adenosine was infused at a rate of 140 μ g/kg per min, and the average peak velocity increased to 57 cm/s. *APV*, Average peak velocity; *d*, diastole; *DSVR*, diastolic systolic velocity ratio; *s*, systole.

other than the status of the epicardial coronary artery can cause abnormalities in the Doppler-derived coronary flow reserve. A failure to increase the APV in response to adenosine may be due to an unresponsive microvasculature rather than the presence of a significant stenosis. Conditions associated with high resting velocities (i.e., left ventricular hypertrophy) will not augment flow further with adenosine. This makes the technique less useful to assess the hemodynamic

significance of a stenosis in the presence of conditions associated with microvascular abnormalities, such as a previous myocardial infarction, heart failure, left ventricular hypertrophy, and, perhaps, diabetes mellitus.

THERMODILUTION-DERIVED ABSOLUTE FLOW AND ABSOLUTE FLOW RESERVE

Thermodilution-derived methods of measuring absolute flow and flow reserve overcome some of the shortcomings of Doppler-derived absolute flow reserve. Advances in catheter and wire technology have allowed for the collection of thermodilution data using intracoronary saline at room temperature. The continuous infusion of room-temperature saline induces maximal hyperemia, which allows for direct quantification of absolute coronary flow and resistance. This method has been shown to be reliable, reproducible, operator independent, and does not require the administration of adenosine. Thermodilution flow reserve methodology has also been used to calculate coronary resistance, described by the index of coronary microvascular resistance or IMR. IMR values >40 are considered elevated and predict the incidence of 30-day major adverse cardiac events, infarction size, and the degree of microvascular obstruction in patients undergoing percutaneous coronary intervention (PCI). Specific details regarding thermodilution methods have been reviewed elsewhere.⁵ The thermodilution methods are primarily used as research tools and have not been embraced clinically.

PRESSURE-DERIVED FRACTIONAL FLOW RESERVE

FFR is an alternative method of assessing the hemodynamic significance of an intermediate or ambiguous coronary stenosis. This popular technique is simple to perform and is based on the measurement of the coronary pressure, using an angioplasty guidewire outfitted with a micromanometer. The mathematics and experimental basis of this technique have been described.⁶ The underlying concept states that FFR is the ratio between the maximum achievable blood flow in the presence of a stenosis and the theoretical maximum flow in the absence of a stenosis (i.e., a normal artery). It can be calculated simply by using the formula:

$$FFR = \frac{\text{Mean hyperemic distal intracoronary pressure}}{\text{Mean hyperemic aortic pressure}}$$

The unequivocal normal value is 1.0.

The derivation of this formula is simple (Fig. 14.8). Based on Ohm's law, coronary flow is equal to the coronary driving pressure divided by coronary resistance. Coronary driving pressure can be expressed as the difference between coronary artery pressure and coronary venous pressure (P_v), with P_v best approximated by the right atrial pressure. In the absence of a coronary stenosis, the coronary driving pressure equals the difference between the aortic pressure (P_a) and P_v . Thus flow in the absence of a stenosis is simply $(P_a - P_v)/\text{Resistance 1}$. Similarly, in the presence of a coronary stenosis, the coronary driving pressure equals the difference between the pressure distal to the stenosis (P_d) and P_v , with flow in the presence of a stenosis expressed as $(P_d - P_v)/\text{Resistance 2}$. FFR is the ratio of maximum flow in the presence of a stenosis to the theoretical maximum flow if the artery were normal. Therefore the formula becomes:

$$FFR = \frac{(P_d - P_v)/R_2 \text{ under maximal hyperemia}}{(P_a - P_v)/R_1 \text{ under maximal hyperemia}}$$

During maximum hyperemia, because the resistances R_1 and R_2 are very low, they will be similar and cancel out. In addition, P_v is usually very low, does not contribute significantly, and is usually eliminated from the calculation. The formula becomes simplified as:

$$FFR = \frac{\text{Mean hyperemic } P_d}{\text{Mean hyperemic } P_a}$$

Although the normal value of 1.0 is well accepted and has been firmly established in humans,⁷ the value below which a stenosis is deemed *significant* is of some debate. Several investigations have determined the FFR value associated with ischemia on noninvasive tests.^{8–10} These investigations involved patients with single vessel disease, stable chest pain syndromes, and moderate coronary lesions and excluded patients with potentially confounding conditions such as left ventricular hypertrophy or prior myocardial infarction. Based on these studies, lesions associated with ischemia have an FFR of <0.75. In one of the original validation studies of FFR (using an FFR cutoff of <0.75 as significant) against noninvasive ischemic testing, the sensitivity of FFR in the identification of reversible ischemia was 88%, the specificity 100%, the positive predictive value 100%, the negative predictive value 88%, and the accuracy 93%¹⁰. Lesions with an FFR between 0.75 and 0.80 are generally recognized as *borderline* and may, in fact, represent significant lesions, particularly in the setting of left ventricular hypertrophy or high right atrial pressure. In these

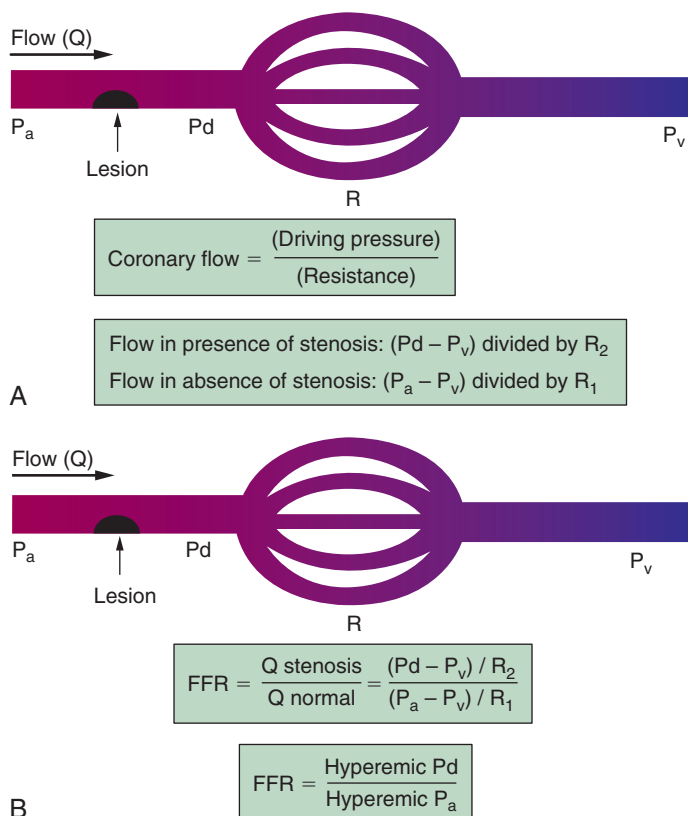


Fig. 14.8 (A and B) Derivation of fractional flow reserve. *FFR*, Fractional flow reserve; P_a , aortic pressure; P_v , coronary venous pressure or right atrial pressure; P_d , coronary pressure distal to the stenosis; Q , coronary blood flow; R , resistance.

borderline or greyzone cases, the decision whether or not to revascularize was typically left to the discretion of the operator. However, studies show a gradient of risk within the borderzone values and currently, it is generally accepted to revascularize lesions with an *FFR* of <0.80 to allow for error and account for the chance that these borderline values may be ischemic. This cutoff value has served as the basis of many important large-scale clinical trials using *FFR* for revascularization decisions.

FRACTIONAL FLOW RESERVE TECHNIQUE

Assessment of *FFR* is fairly straightforward to perform, using basic interventional skills (Fig. 14.9). After engaging a standard angioplasty guide catheter into the coronary ostium, the pressure wire is zeroed and calibrated. Because the procedure involves instrumentation of the coronary artery, the University of Virginia Cardiac Catheterization Laboratory (Charlottesville, Virginia) typically administers an intravenous bolus of unfractionated heparin (50 U/kg) before wire passage distal to the coronary lesion to prevent catheter and wire thrombosis. Once the operator has established that the pressures recorded from the catheter tip and the pressure wire's micromanometer are identical (Fig. 14.9B and C), the wire is advanced down the coronary artery and the transducer positioned several millimeters distal to the lesion in question. The pressure waveforms are recorded at rest and during maximal hyperemia induced by one of several methods.^{11–15} Intracoronary adenosine is perhaps the simplest method. During the early trials and clinical experiences with *FFR*, clinicians typically used 12 to 40 μg injected as a bolus into the right coronary artery and 16 to 60 μg into the left coronary artery using a concentration of 10 $\mu\text{g}/\text{mL}$. Additional studies learned that higher doses are necessary to achieve maximal hyperemia with doses of up to 600 μg needed to achieve the maximal hyperemia observed with an intravenous infusion of adenosine.^{12–14} Side effects, such as transient high-grade atrioventricular block, bradycardia, and asystole, become more common with high-dose intracoronary bolus injections. Thus the optimal intracoronary

dose balances adequate hyperemia with minimal side effects. A study by Adjedj et al.¹⁵ found that the optimal dose was 100 μg for the right coronary artery and 200 μg for the left coronary artery to provide an FFR value within 0.01 of the value associated with maximal hyperemia and with minimal side effects. To capture maximal hyperemia, simultaneous catheter tip (or P_a) and distal wire pressure (P_d) are recorded for 15 to 20 seconds. The point of maximum hyperemia is apparent at the nadir of the mean pressure typically observed 10 to 30 seconds after the bolus is administered.

An infusion of adenosine through a large peripheral vein at 140 $\mu\text{g}/\text{kg}/\text{min}$ is an alternative and commonly used method of achieving hyperemia. Intravenous administration provides a longer period of hyperemia and a more stable response, but at a greater cost and more potential for systemic side effects such as flushing, dyspnea, hypotension, and chest pain. Adenosine is contraindicated in patients with active bronchospasm and should be used with caution in those with obstructive lung disease. Using this methodology, hyperemia is usually maximal 1 to 2 minutes into the peripheral infusion. Because of the short half-life of adenosine, it was believed that intravenous infusions should be administered through a large central vein such as the femoral vein; however, it appears that hand-vein infusion of adenosine results in the same FFR values as central femoral-vein infusion.¹⁶ Again, similar to the intracoronary bolus of adenosine, the point of maximum hyperemia with intravenous infusion is at the nadir of the mean pressure typically observed 1 to 2 minutes after the start of the infusion. Fluctuations in the extent of hyperemia during the course of the infusion are sometimes observed. The mechanism of these

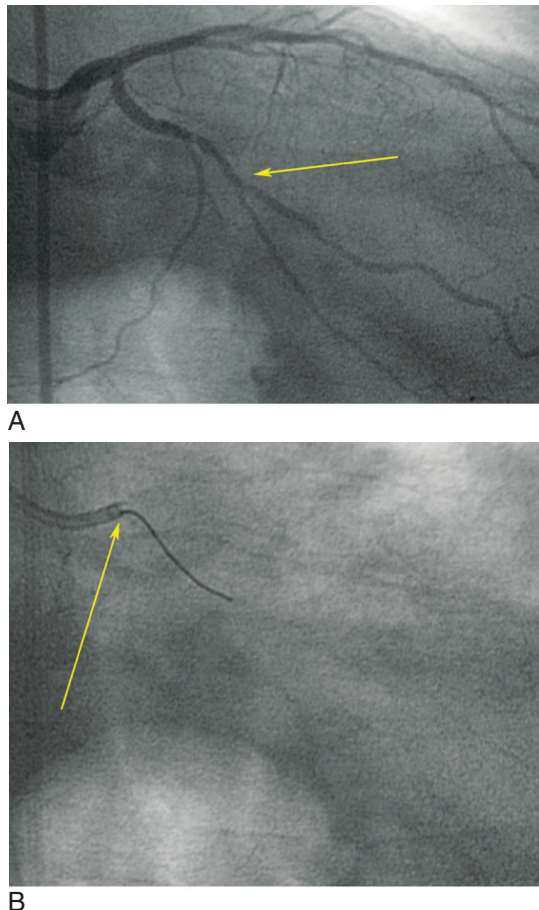


Fig. 14.9 (A) Example of the method for measuring fractional flow reserve. A moderate lesion is seen in the midportion of the left circumflex artery (arrow). (B) The pressure wire is placed initially with the transducer (arrow) at the guide tip to ensure that Fig. 14.9

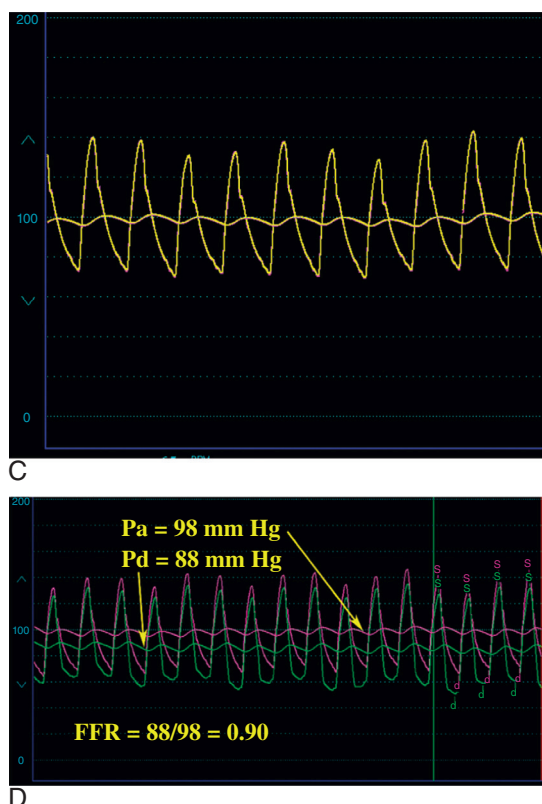


Fig. 14.10, cont'd (C) the guide pressure and wire transducer pressure are identical. The wire is then advanced with the transducer positioned past the lesion. Baseline pressures are usually sampled and then maximal hyperemia induced with adenosine. (D) During maximal hyperemia, pressures are sampled at the catheter tip or aorta and distal to the lesion. In this case the $FFR = 0.90$. *FFR*, Fractional flow reserve; P_a , aortic pressure; P_d , coronary pressure distal to the stenosis.

fluctuations is not entirely clear and may represent a compensatory response to hyperemic flow. If observed, the “true” FFR is the lowest value observed.¹⁷

Regardless of the method used to achieve hyperemia, the pressure waves are recorded during maximal hyperemia, and FFR is simply calculated as P_d/P_a , as described above (Fig. 14.9D). Most operators end the procedure by withdrawing the pressure transducer to the catheter tip, verifying that the pressures remain equal, and confirming that transducer drift has not occurred, establishing the reliability of the measurements.

Several important considerations and potential sources of errors when making these measurements include the following:

1. The operator should be careful to note if the engagement of the guide catheter causes pressure damping or ventricularization. This will falsely lower the P_d and cause an erroneously high calculated FFR (Fig. 14.10). Furthermore, if the intracoronary method of adenosine administration is used, the operator should avoid using a guide catheter with side holes and be sure the catheter is selectively engaged in the artery. This ensures that the adenosine bolus actually enters the coronary circulation; otherwise, maximal hyperemia will not occur, underestimating the FFR.
2. If the FFR value obtained with intracoronary adenosine is in the borderline range (i.e., 0.80–0.81), the operator might consider repeating the measurement at higher bolus doses of adenosine (up to 600 μ g) or switch to intravenous adenosine before concluding that the lesion is not hemodynamically significant.
3. Operators should be aware that theophylline and related drugs block adenosine’s activity and may blunt the hyperemic response, falsely elevating the FFR. Patients should hold these medications for at least 12 hours to avoid this pitfall. One cup of coffee more than 1 hour before adenosine administration does not interfere with measurements.¹⁸



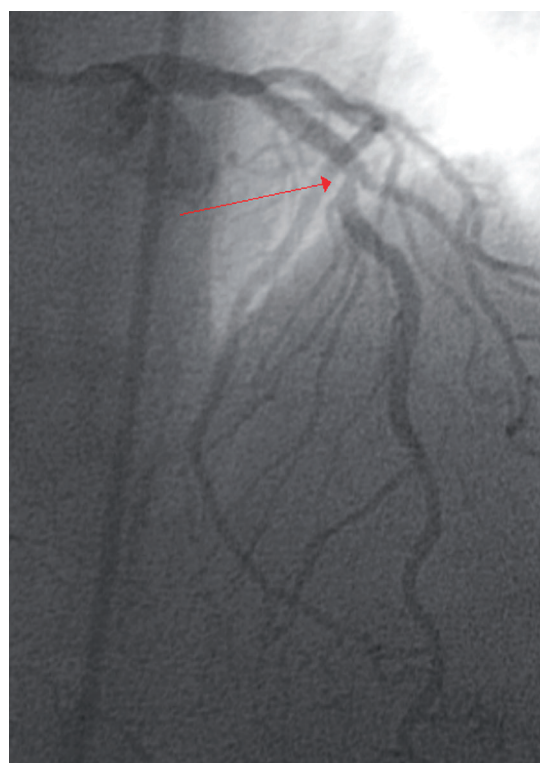
Fig. 14.10 Example of a potential source of error during fractional flow reserve measurement. (A) There is subtle ventricularization of the guide catheter pressure waveform, resulting in a falsely low aortic pressure (74 mm Hg) with a mean coronary pressure distal to the stenosis of 68 mm Hg and a calculated fractional flow reserve of 0.92, suggesting that this lesion is not significant. (B) However, when the guide catheter is repositioned into the aorta, the mean aortic pressure rises to 86 mm Hg with a mean coronary pressure distal to the stenosis of 64 mm Hg and a calculated fractional flow reserve of 0.75, suggesting that the lesion is, in fact, a significant one.

4. A point often not considered relates to the potential risk of guidewire instrumentation of an atherosclerotic coronary artery. Rarely, the guidewire passage disrupts or dissects the atherosclerotic plaque and causes abrupt vessel closure. Therefore if the lesion in question is not amenable to percutaneous interventional techniques, FFR should not be performed.
5. Finally, transducer pressure drift can occur during the performance of the procedure. For this reason, confirmation that the catheter-tip pressure and the pressure wire transducers are the same at the conclusion of the case provides the operator with greater confidence in the FFR measurement obtained.

CLINICAL APPLICATIONS AND PROGNOSTIC VALUE OF FRACTIONAL FLOW RESERVE

Coronary hemodynamic assessment is indispensable in the current practice of interventional cardiology. FFR is most commonly used to assess lesions that appear only modestly stenotic. Angiography alone is notoriously misleading in this subset. In fact, even experienced interventional cardiologists are often unable to discriminate significantly from nonflow limiting lesions by angiography alone, and they often disagree regarding the significance of the same lesions!¹⁹ In general, when a cardiologist stares at an angiogram over and over again, scratches their head and requests multiple additional views, and still remains unsure, it is time to perform FFR.

Fig. 14.11 is an example of the utility of FFR in a 52-year-old female with peripheral vascular disease and a remote history of a coronary intervention on the right coronary artery, now presenting with an anginal chest pain syndrome. Angiography revealed total occlusion of a small right coronary artery and normal-appearing left main and left circumflex (LCX) arteries. The left anterior descending (LAD) artery had a stenosis of moderate severity at the



A



B

Fig. 14.11 (A) Example of the value of fractional flow reserve in determining the significance of a moderate stenosis of the left anterior descending artery (*arrow*) in a 52-year-old female. (B) The fractional flow reserve was calculated at 0.71, consistent with a hemodynamically significant stenosis.

bifurcation of a diagonal branch (Fig. 14.11A). Despite imaging in multiple views, the lesion did not appear particularly worrisome or severe. However, FFR assessment demonstrated that this lesion was hemodynamically significant with an FFR calculated at 0.71 (Fig. 14.11B).

Another common indication for FFR is the assessment of ambiguous lesions, that is, coronary disease not well visualized by angiography either because of overlapping segments, vessel tortuosity, or lesion eccentricity. FFR assessment is especially valuable when the patient's clinical symptoms are atypical. The FFR measurement can be extremely valuable to the cardiologist who wishes to reconcile the clinical syndrome with the angiogram. If the ambiguous lesion

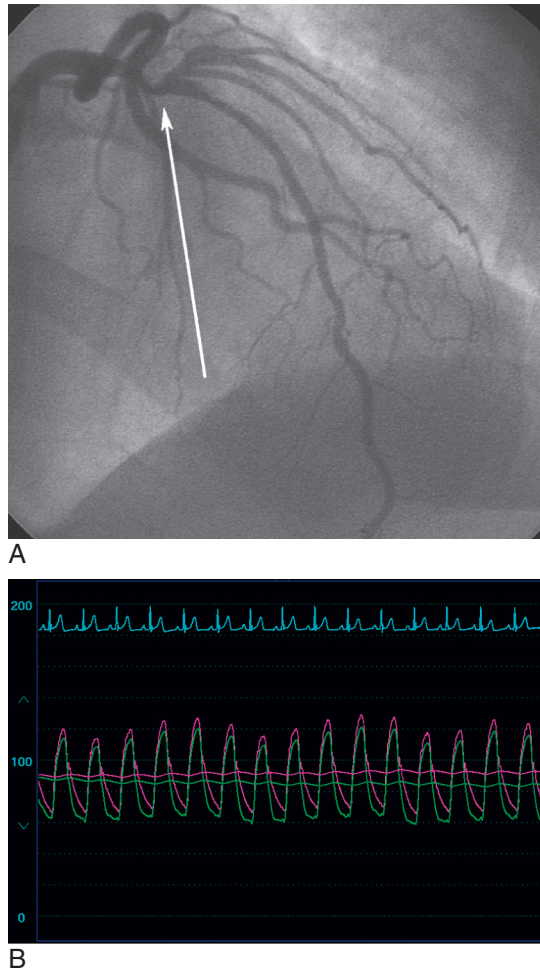


Fig. 14.12 (A) Example of an ambiguous lesion in the proximal left anterior descending artery. Note the complex array of vessels that surround the lesion, making it difficult to see by angiography (*arrow*). (B) Fractional flow reserve was 0.94, confirming that this lesion was not significant.

is not significant, it is unlikely that the patient's atypical symptoms are because of the stenosis or that the condition will improve with revascularization. Alternatively, the angiogram may underrepresent an important stenosis, causing the practitioner to erroneously dismiss the symptoms as noncardiac.

The angiogram shown in Fig. 14.12 reveals an ambiguous lesion in the proximal LAD artery in a 55-year-old morbidly obese female with dyspnea on exertion and an abnormal, but nondiagnostic, noninvasive study. Left ventricular function was normal, and no atherosclerotic disease was found in the right coronary and circumflex arteries. In this case a complex array of branches emanated from the proximal segment of the LAD artery, overlapping the area in question. In addition, the patient's body habitus limited the range of angiographic angles, preventing optimal visualization of the area. An FFR of 0.94 confirmed that this lesion was not significant and, therefore, not responsible for the clinical syndrome.

It is important for the cardiologist to understand the natural history of patients with ambiguous lesions or lesions of moderate severity in whom revascularization is deferred based on the presence of a "nonischemic" FFR. Soon after the method of FFR was described, several investigators explored this and found that when the FFR was >0.75 and revascularization deferred, the 12-month rate of major adverse cardiac events was low (around 10%).^{20–23} Few events during follow-up were nearly all caused by target vessel revascularization; the rates of death or nonfatal infarction were very low. Long-term follow-up studies and more recent investigations performed in the era of drug-eluting stents have

solidified and confirmed these findings. The deferral of revascularization for lesions with an FFR of >0.75 is safe, with very low event rates over 5 years observed in the DEFER trial, where the rate of cardiac death or myocardial infarction related to the deferred stenosis was found to be $<1\%$ per year.²⁴ Subsequent follow-up out to 15 years of the DEFER trial patients continued to show no significant difference in death rate between the two groups (33.0% vs. 31.1%, $P = .79$) but lower rates of myocardial infarction in the deferred PCI group (2.2% vs. 10.0%, $P = .03$).²⁵ In the FFR Versus Angiography for Multivessel Evaluation (FAME) trial, 1005 patients with multivessel disease were randomized to angiographic-guided versus FFR-guided coronary intervention.²⁶ For those randomized to an FFR-guided strategy, PCI was performed only for lesions with an FFR of less than or equal to 0.80. The FFR-guided strategy was associated with fewer lesions requiring treatment and less use of stents. Importantly, at 2 years the rate of death and myocardial infarction in the FFR-guided group was significantly lower than the angiographic-guided group (8.4% vs. 12.9%, $P = .02$). For deferred lesions at 2-year follow-up, the rate of myocardial infarction related to the lesion was only 0.2%, and the rate of revascularization was only 3.2%. At 5-year follow-up, the FFR-guided PCI group was treated with fewer stents, and the risk of major adverse cardiac events remained similar in both the angiography-guided PCI and FFR-guided PCI groups confirming the safety of this strategy.²⁷

The prognostic value of FFR has been explored in a large meta-analysis consisting of more than 6000 patients.²⁸ An important finding of this analysis is that in terms of prognostic value, FFR should be considered as a continuum rather than as a binary value. The lower the value of FFR, the greater the benefit from revascularization; in this large-scale meta-analysis, the threshold for benefit is an FFR of <0.75 to 0.80. This large analysis also found that FFR measured after coronary intervention also carries important prognostic value, and the lower the FFR after intervention, the worse the outcome, likely reflecting the presence of residual, untreated disease.

Finally, there has been little data regarding the fate of lesions with an FFR of less than or equal to 0.80 treated medically. In FAME-2 nearly 900 patients under consideration for coronary intervention underwent FFR and were randomized to intervention of all lesions with an FFR of less than or equal to 0.80 ($n = 447$) versus optimal medical therapy alone ($n = 441$).^{29,30} The study was halted early because the event rate in the medical therapy group was noted to be significantly higher than in the intervention group. The composite primary endpoint of death, myocardial infarction, or urgent revascularization at 12 months was 4.3% in the intervention group and 12.7% in the medical therapy group ($P < .001$); this was mostly due to the higher rate of urgent revascularization procedures in the medical therapy group (11.1% vs. 1.6%, $P < .001$).²⁹ At 5 years, the primary endpoint remained lower in the PCI group and patients with FFR >0.80 continued to do well with medical therapy.³⁰ Not surprisingly, in the FAME-2 study, the lesions associated with adverse cardiac events during follow-up had a lower FFR (median FFR = 0.68) compared to those that had no event during follow-up (median FFR = 0.80).³¹

The data from these clinical trials, as well as extensive data now available from large registries,^{32–35} confirm the benefit of FFR-guided PCI and reassure the practitioner and the patient that revascularization is not necessary for lesions that appear moderately narrowed on angiography with an FFR of >0.80 thus emphasizing the importance of revascularizing lesions that are functionally significant, regardless of their angiographic appearance.

FRACTIONAL FLOW RESERVE IN SPECIFIC LESION AND PATIENT SUBSETS

Questions are often raised regarding the value and validity of FFR in specific lesion and patient subsets, such as patients with acute coronary syndromes, left main stem disease, side branch lesions, multivessel disease, tandem or serial stenoses, and saphenous vein graft lesions.

FRACTIONAL FLOW RESERVE IN ACUTE CORONARY SYNDROMES

The studies demonstrating favorable outcomes when revascularization is deferred in patients with nonischemic FFR consisted primarily of patients with stable coronary disease. Do these findings apply to patients with acute coronary syndromes (non-ST segment elevation myocardial infarction [NSTEMI], and unstable angina) who are found to have moderate stenosis? The unstable plaque responsible for an acute coronary syndrome might behave less predictably and have a different natural history than a stable plaque with the same extent of luminal narrowing and FFR. The theory underlying FFR assumes a stable and minimal microvascular resistance, which may not be the case in patients with acute coronary syndrome. However, multiple studies have shown FFR to be reliable in patients with NSTEMI and unstable angina.^{23,36,37} FFR is also useful and reliable in determining the significance of “nonculprit” lesions found on angiography in patients undergoing emergent angiography for NSTEMI.³⁸ The role of FFR in the assessment of the non-culprit lesion in patients with STEMI is somewhat unclear. Although one study found that FFR predicted outcome in the noninfarct related artery in patients with STEMI,³⁹ additional studies have failed to demonstrate a benefit of FFR-guided therapy over angiography-guided therapy for nonculprit lesions found at the time of STEMI hinting that STEMI may disturb coronary physiology in other vascular beds leading to inaccuracies in FFR.^{40,41}

FRACTIONAL FLOW RESERVE IN LEFT MAIN DISEASE

Atherosclerotic narrowing of the left main coronary artery is a potentially life-threatening condition but may not always be imaged clearly by coronary angiography.⁴² This is particularly true for lesions that involve the ostium, an area commonly obscured from plain view. These lesions are ideal for assessment by FFR, allowing the physician fewer sleepless

nights worrying about a decision based solely on an imperfect angiogram. The value of FFR in the assessment of ambiguous left main lesions has been explored, and several studies have demonstrated the safety of deferring revascularization of ambiguous left main lesions with a nonischemic FFR.⁴³⁻⁴⁶

Isolated left main disease is rare, and very often there is coexisting disease in the LAD and/or LCX arteries. In the presence of coexisting disease, the wire is generally placed in the nondiseased vessel; for example, if there is severe LAD disease, the wire is placed in the LCX. However, concern has been raised that the existence of “downstream” disease may cause adequate augmentation of blood flow across the left main during hyperemia, rendering FFR inaccurate by falsely elevating the FFR. This question has been addressed experimentally and in humans. It appears that although downstream disease does affect FFR measurement, the effect is small and clinically not important unless the disease is very severe.^{47,48} Examples of ambiguous left main lesions assessed by FFR are shown in Figs. 14.13 and 14.14.

FRACTIONAL FLOW RESERVE FOR THE ASSESSMENT OF BIFURCATION AND SIDE BRANCHES

It is often very difficult to determine the significance of a stenosis of disease by angiography alone. This is particularly true after stenting the main branch of a bifurcation; the side branch often appears “pinched” and may lead to unnecessary treatment. These lesions can be well assessed by FFR. In general, there is no relation between the angiographic severity of side branch ostial lesions and FFR, and only about 20% of “jailed” side branches are found to be functionally significant despite appearing angiographically severe.⁴⁹

FRACTIONAL FLOW RESERVE IN MULTIVESSEL CORONARY ARTERY DISEASE

Patients with multivessel disease represent another challenging subset. The choice of revascularization (i.e., bypass surgery versus percutaneous intervention) is often based on the angiogram. While a “culprit” lesion is often readily apparent, the significance of disease in other arteries may be difficult to discern. Myocardial perfusion imaging, based on the concept of relative flow reserve, may be entirely normal despite severe three-vessel disease because of “balanced” ischemia.^{50,51} More commonly, however, in the setting of multivessel coronary disease, myocardial perfusion imaging generally detects only the most severe stenosis and cannot be used to determine the significance of another, ambiguous, lesion.^{52,53} Again, in such scenarios FFR can allow the physician to make revascularization decisions with great confidence (Fig. 14.15). The clinical value of FFR and outcomes when used in decision-making in patients with multivessel disease have been established in the FAME trial, discussed earlier.²⁶

An important finding from the FAME trial was the continued demonstration of the poor correlation between the angiographic percentage of stenosis of a lesion and its functional significance. In this large cohort of patients with multivessel disease, all lesions with at least 50% stenosis were assessed by FFR. Similar to other studies, only 35% of lesions categorized as “intermediate” with 50% to 70% stenosis by angiography had an FFR of <0.80 .⁵⁴ Perhaps more interesting was the observation that 20% of lesions characterized as “severe” with 71% to 90% stenosis were found to be not significant by FFR.⁵⁴ When choosing patients for FFR assessment, there are several additional variables to consider beyond the percentage of stenosis. Arteries with long lesions or small absolute minimal luminal diameter and those that subtend a large area of myocardium are more likely to be hemodynamically significant by FFR.^{55,56}

Most of the data have focused on FFR-guided PCI, but many patients with multivessel disease require bypass surgery. A recent meta-analysis summarized the available data for FFR-guided CABG.⁵⁷ This analysis found that FFR guidance reduced the number of conduits used and simplified the operation but there was no difference in adverse events at 6 and 12 months in FFR-guided CABG compared to angiography-guided CABG. However, bypass grafts anastomosed to coronary arteries with abnormal FFR values were more likely to remain patent compared to those anastomosed to arteries with nonischemic FFR values, especially if the graft is arterial rather than a venous graft.

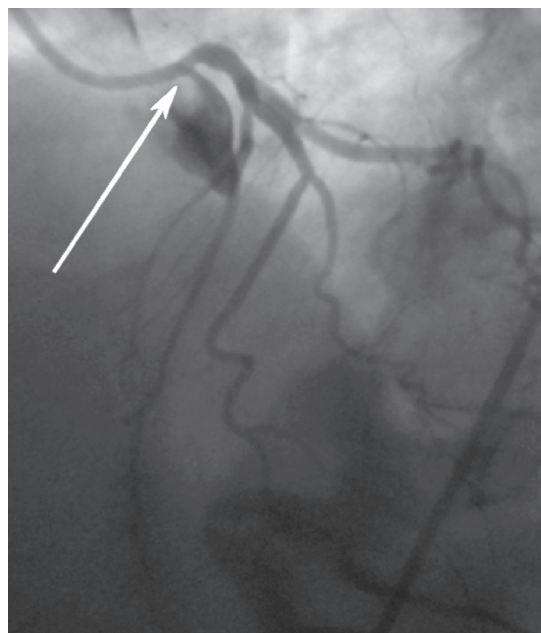
FRACTIONAL FLOW RESERVE IN DIFFUSE DISEASE AND FOR SERIAL OR TANDEM LESIONS

Diffuse coronary disease and serial or tandem lesions in the same artery represent unique subsets difficult to assess by angiography alone. Coronary hemodynamic assessment can prove extremely valuable in these settings. Recall that coronary angiography represents only the arterial lumen. Angiography accurately identifies and characterizes coronary artery disease only in the presence of a normal *reference* segment. Atherosclerosis may involve the entire artery, however, making it difficult to appreciate the extent of arterial narrowing by angiography alone. Coronary hemodynamic assessment has the potential of detecting diseases not readily apparent on angiography. In a study of patients with angiographic coronary disease in one artery, a significant pressure gradient existed between the aorta and the distal coronary artery of another, angiographically “normal” artery, in 57% of arteries; in 8% of such normal arteries, the FFR was <0.75 , demonstrating the truly diffuse nature of coronary disease.⁵⁸

A modification in the technique for FFR assessment is required in the setting of diffuse coronary disease. Because there is no specific, focal lesion, the pressure transducer is positioned in the distal artery and slowly pulled back during hyperemia. This technique requires sustained hyperemia using either an intravenous infusion of adenosine at 140 $\mu\text{g}/\text{kg}$ per min or a 0.4 mg bolus of regadenoson. The duration of hyperemia induced by intracoronary adenosine is too brief and will likely dissipate during the course of the pullback. The lowest FFR value is at the most distal site, and a slow, progressive rise in the FFR occurs as the pressure wire is withdrawn toward the guide catheter (Fig. 14.16).



A



B

Fig. 14.13 Example of the role of fractional flow reserve (FFR) determination in the assessment of an ambiguous left main lesion. (A) The catheter pressure waveform ventricularized and damped. (B) Angiography identified an ostial narrowing of moderate severity of the left main artery (*arrow*).

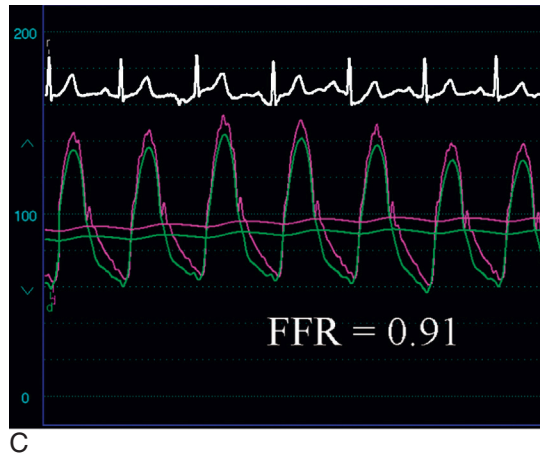


Fig. 14.13, cont'd. (C) Coronary pressure measurement determined the fractional flow reserve at 0.91.

Occasionally the operator may note an abrupt step-up in the hyperemic pressure gradient, suggesting a more focal area of disease treatable with stenting.

Determination of the significance of tandem or serial lesions within the same artery represents a significant challenge to the operator. One cannot simply calculate hyperemic P_d/P_a at each site because significant interaction occurs between the two lesions. With two lesions in tandem, the FFR of the distal lesion is influenced by both proximal and distal stenosis. Similarly FFR of the proximal stenosis is affected by the presence of the distal stenosis. Flow through the first stenosis will be submaximal, leading to a falsely high FFR, potentially misclassifying a hemodynamically significant stenosis. Equations for the calculation of FFR for sequential stenoses have been described but are cumbersome and not suitable for application to routine clinical practice. The usual method is to use the pullback technique with the FFR beginning across the distal stenosis. The lesion site with the largest step-up in pressure (i.e., the most severe lesion) is treated first, and then FFR is repeated for the remaining lesions. This method has been shown to be safe and effective with excellent clinical outcomes.⁵⁹ The nonhyperemic indices, to be described later in this chapter, may also be useful for classifying the significance of tandem lesions. Fig. 14.17 shows the use of FFR in a case of sequential stenoses that involve the LAD artery.

FRACTIONAL FLOW RESERVE FOR THE ASSESSMENT OF SAPHENOUS VEIN GRAFT LESIONS

We now have established that deferral of revascularization for lesions with nonischemic FFR involving native coronary arteries is safe and associated with a very low event rate. Can we extrapolate that information to saphenous vein graft lesions? Studies focusing on outcomes of FFR-based deferral in vein graft lesions found high rates of adverse cardiac events when deferring intervention on a vein graft lesion despite nonischemic FFR.^{60,61} This can most likely be explained by a different natural history for vein graft lesions as compared to native coronary artery disease.

LIMITATIONS OF FRACTIONAL FLOW RESERVE THEORY

FFR assessment has become an established adjunctive to angiography, and FFR values ≤ 0.80 are generally accepted as hemodynamically significant or *ischemic*. Nevertheless, several important limitations and criticisms exist.⁶² Great care should be taken when obtaining these FFR measurements. The method relies on achieving maximal hyperemia; if this is not achieved because of an error in infusion of adenosine, the FFR will be falsely high and the lesion severity underestimated. Similarly, abnormalities of the microvasculature, as may be seen in patients with extensive infarction or severe ventricular hypertrophy, may impair hyperemic flow, jeopardizing the validity of FFR. The guide catheter may interfere with blood flow, so carefully review for evidence of damping and consider backing out the guide during the infusion. Some of the underlying assumptions of FFR theory may be in error. For example, myocardial resistance may not be constant or uniform during hyperemia. In fact, microvascular resistance may increase from rest to hyperemia in the presence of a stenosis. Thus canceling resistance may overestimate the true FFR. As previously discussed earlier, a common shortcut is to ignore venous pressure; however, this may not be appropriate, particularly if the FFR is close to an ischemic cutoff. Consider the following example: If, during maximum hyperemia, $P_a = 105$ mm Hg, $P_d = 86$ mm Hg, and the right atrial pressure is ignored, then $FFR = 0.82$, and the lesion is classified as *nonsignificant*. If right atrial pressure were taken into consideration and measured only 5 mm Hg, then the corrected FFR would not change the designation of the lesion ($FFR = 0.81$). However, if the right atrial pressure were elevated at 20 mm Hg, then the FFR would

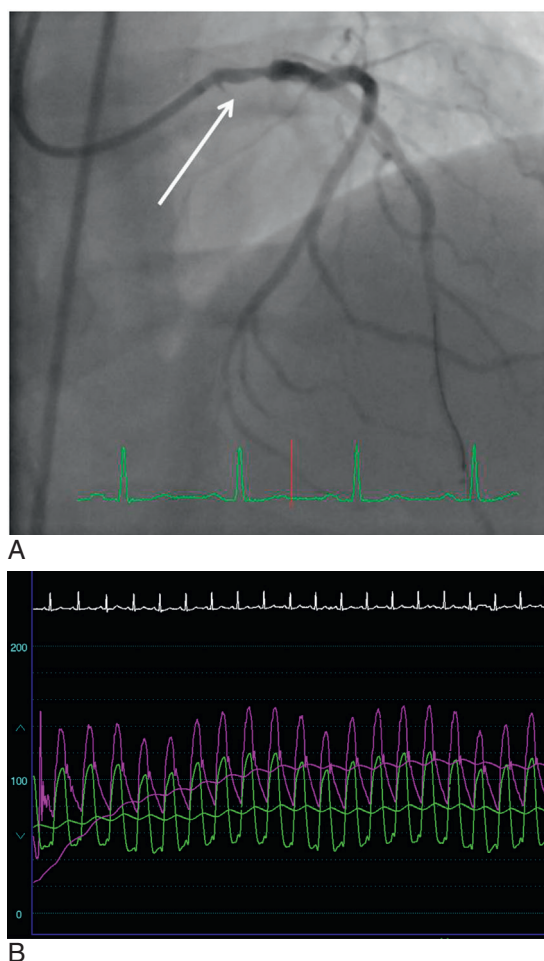


Fig. 14.14 Example of the use of fractional flow reserve for the assessment of left main coronary artery disease in a 58-year-old male with angina pectoris. (A) Angiography of the left coronary artery shows narrowing of the left main stem of unclear significance (*arrow*). (B) With the pressure wire positioned in the left anterior descending artery, the fractional flow reserve was measured and found to be 0.68, which showed that the left main lesion was hemodynamically significant.

now be calculate to 0.78, changing the lesion designation. Therefore it may make sense to consider measuring right atrial pressure when lesions are close to significant, particularly if elevated right atrial pressure is suspected. In general, this phenomenon is rare, and routine measurement of right atrial pressure has been found not to be necessary as the effect on the FFR is truly minimal.⁶³

ALTERNATIVES TO TRADITIONAL FFR

In an effort to simplify the procedure and avoid the potential side effects and cost of adenosine, several alternatives to traditional FFR have been developed.

CONTRAST FFR

The iodinated contrast agents we use to perform coronary angiography cause transient coronary hyperemia. Although hyperemia is not maximal, this property has been advocated as a potential alternative to traditional FFR and is known as contrast FFR (cFFR); it is calculated as the lowest mean P_d/P_a value after the intracoronary injection of a contrast agent. Studies have demonstrated the similarity between cFFR and FFR and a cutoff of <0.83 for cFFR best predicted traditional FFR <0.80 .^{64,65} Hyperemia is very transient, and thus cFFR is not appropriate for pullback analysis due to the

transient hyperemia provided by the contrast media. The risks of its use are similar to those of FFR, with the added risk of increased contrast use.

NONHYPEREMIC PRESSURE RATIOS

Adenosine is associated with side effects, additional cost, and prolonged (albeit minimally) procedure time, which led to efforts to create indices of stenosis severity that do not require inducing hyperemia. These nonhyperemic ratios include

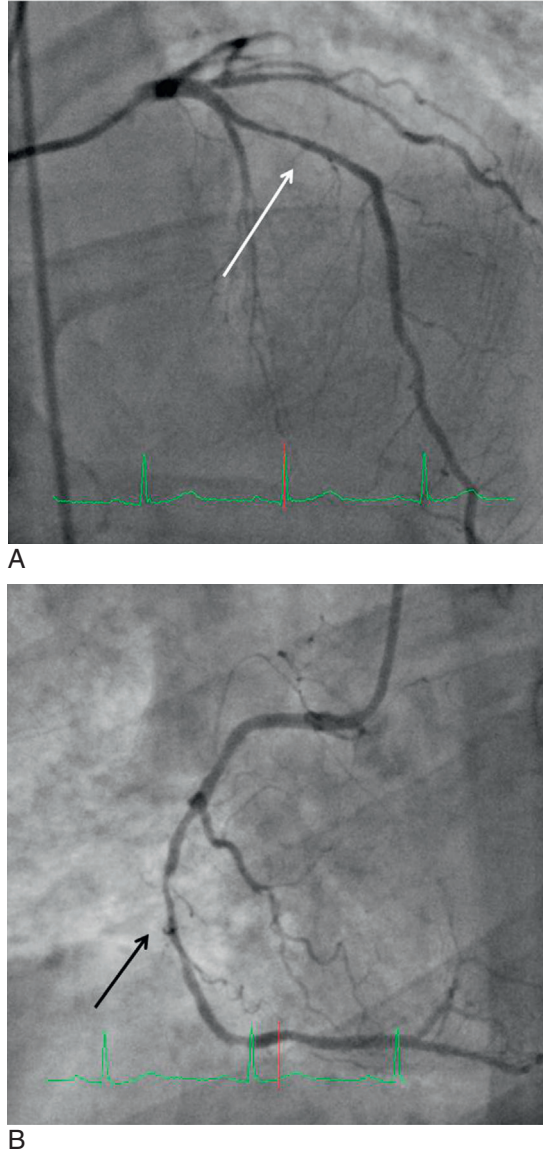


Fig. 14.15 This case is an example of the use of the fractional flow reserve to guide revascularization in multivessel disease. A 60-year-old female with progressive angina is admitted with unstable angina. (A) Angiography of the left coronary artery shows a long stenosis of moderate severity in the left anterior descending artery (*arrow*). (B) Angiography of the right coronary artery revealed a similar long moderate stenosis (*arrow*).

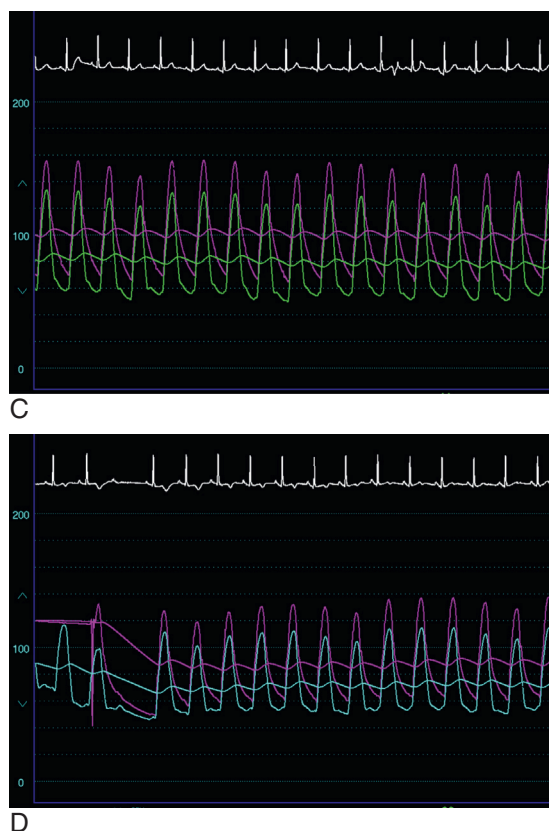


Fig. 14.15, cont'd. (C) Fractional flow reserve of the left anterior descending artery was significant at 0.75. (D) Fractional flow reserve of the right coronary artery was not significant at 0.83. Thus only the left anterior descending artery was stented and remained without angina at 2-year follow-up.

instantaneous wave-free ratio (iFR), resting P_d/P_a , diastolic hyperemia-free ratio (DFR), diastolic pressure ratio (DPR), diastolic pressure ratio (dPR), and resting full cycle ratio (RFR).

INSTANTANEOUS WAVE-FREE RATIO

As noted earlier, the derivation of the formula for FFR relies on adenosine to create maximal hyperemia. During hyperemia, myocardial resistance becomes very low and constant, and thus its contribution is negligible, allowing it to cancel out of the formula. As it turns out, there exists a brief phase during the cardiac cycle known as the *wave-free period*, beginning about a quarter of the way into diastole and ending 5 ms before systole begins, during which microvascular resistance is minimal and constant (Fig. 14.18). Thus the ratio of distal lesion pressure (P_d) to proximal pressure (P_a) during the wave-free period can, in theory, meet the assumptions necessary to calculate the FFR and is known as the iFR.⁶⁶ Coronary resistance is stable during the wave-free period, flow and pressure are related linearly allowing for the calculation of a pressure index without the administration of a vasodilator. Proprietary systems have been developed to generate this index, and this method is easy to perform. When iFR was initially introduced, there was substantial controversy regarding the optimal cutoff value for indicating a significant lesion. Subsequently, a large multicenter trial established the optimal iFR cutoff point for identifying FFR ≤ 0.80 to be 0.90 with an accuracy of ~80%. Thus an iFR value of ≤ 0.89 is considered hemodynamically significant.⁶⁷ It is important to note that an 80% accuracy means that 20% of therapeutic decisions based on iFR would be discordant from FFR-based decisions. The greatest degree of discordance is around the cutoff value. Some advocate that iFR can be used to defer if >0.93 and to revascularize if <0.86 , but lesions with values between 0.87 and 0.92 are in the “gray” zone and should be additionally interrogated with FFR.⁶⁸ Numerous trials have shown that an iFR-guided approach to coronary revascularization was comparable

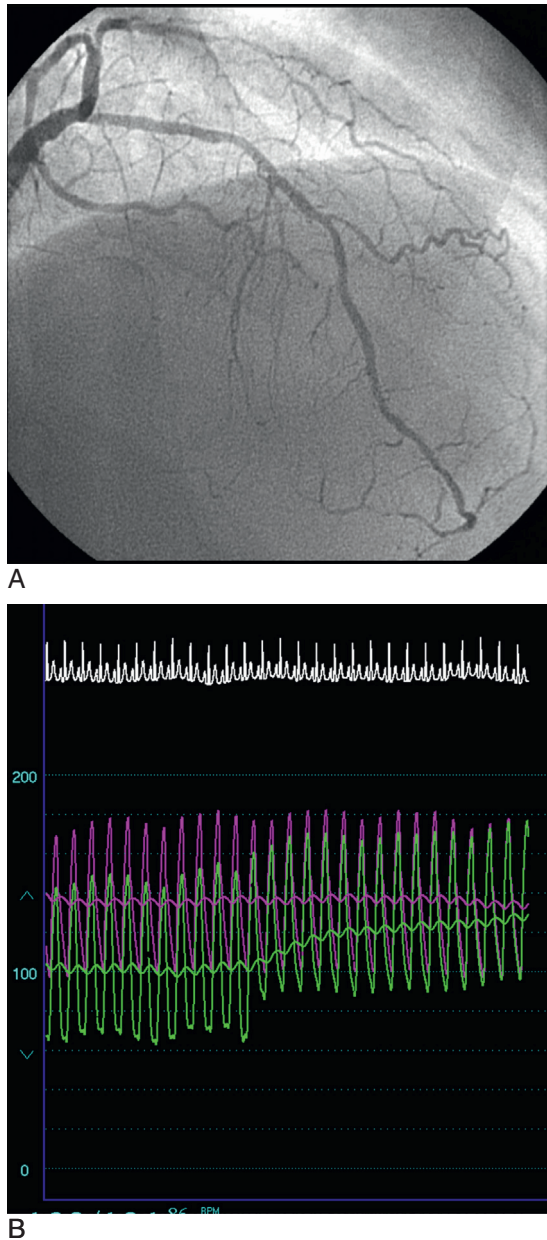


Fig. 14.16 An example of diffuse disease that involves the left anterior descending artery. (A) The angiogram shows a long, moderately diseased segment of the left anterior descending artery. (B) With the pressure wire positioned distally, and under conditions of sustained, maximal hyperemia with intravenous adenosine, the fractional flow reserve was determined to be 0.72. The wire was slowly pulled back, and there was a progressive rise in distal pressure noted throughout the diseased segment all the way to the proximal segment of the artery.

to an FFR-guided approach with no difference in outcomes.^{69,70} This method has also been validated in various lesion subsets, including patients with acute coronary syndrome,⁷¹ left main stem disease,⁷² and serial or tandem lesions.⁷³

RESTING P_D/P_A

Resting P_D/P_A is the simplest of the methods. It is the ratio of the mean P_D and P_A calculated over the entire cardiac cycle. A cutoff of <0.93 for resting P_D/P_A has been suggested and validated against FFR.⁶⁴ However, there is a large

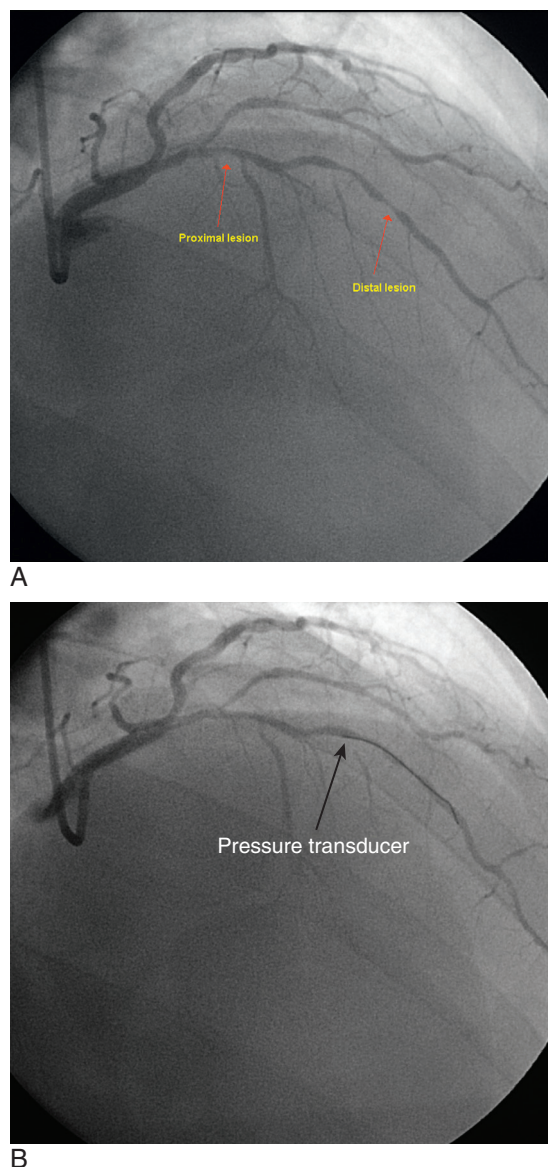


Fig. 14.17 Example of fractional flow reserve in the case of sequential stenoses that involve the left anterior descending artery. (A) An angiographic severe stenosis was noted in the distal portion of the artery with a more proximal, moderate stenosis also present. (B) When the transducer of the pressure wire was first placed across just the proximal, moderate stenosis,

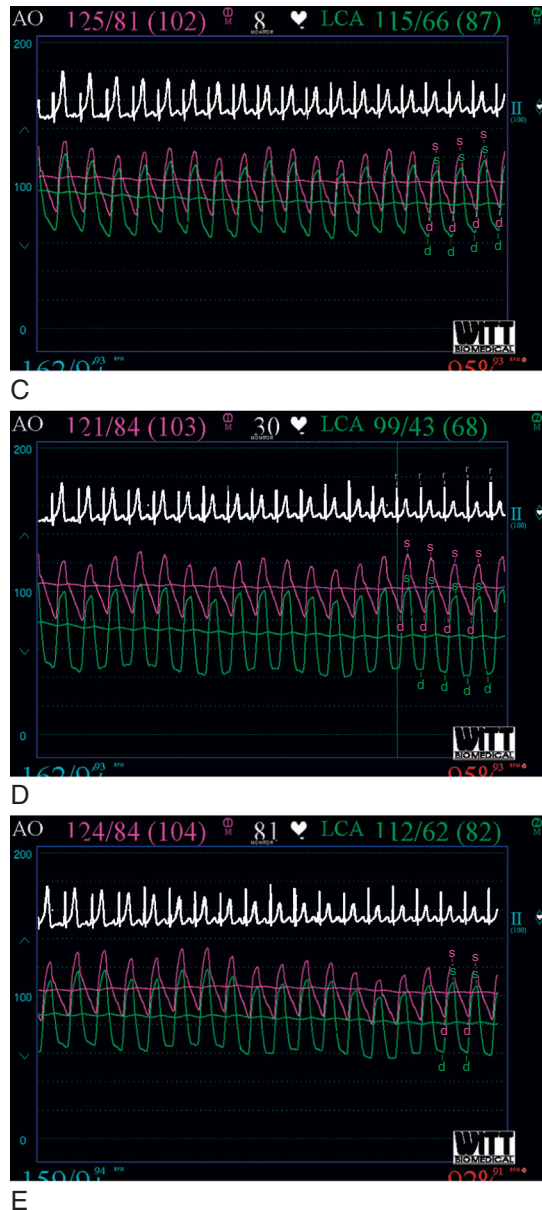


Fig. 14.17, cont'd. (C) the fractional flow reserve was 0.85. (D) When the wire was placed distally across both lesions, the fractional flow reserve was 0.66. The distal lesion was stented successfully, and when the fractional flow reserve was repeated across the proximal lesion, (E) it had now decreased from 0.85 to 0.78. This is an example of how a distal lesion may falsely raise the fractional flow reserve of the more proximal lesion and exemplifies the complexity of the fractional flow reserve assessment in the setting of tandem lesions. *d*, Diastole; *s*, systole.

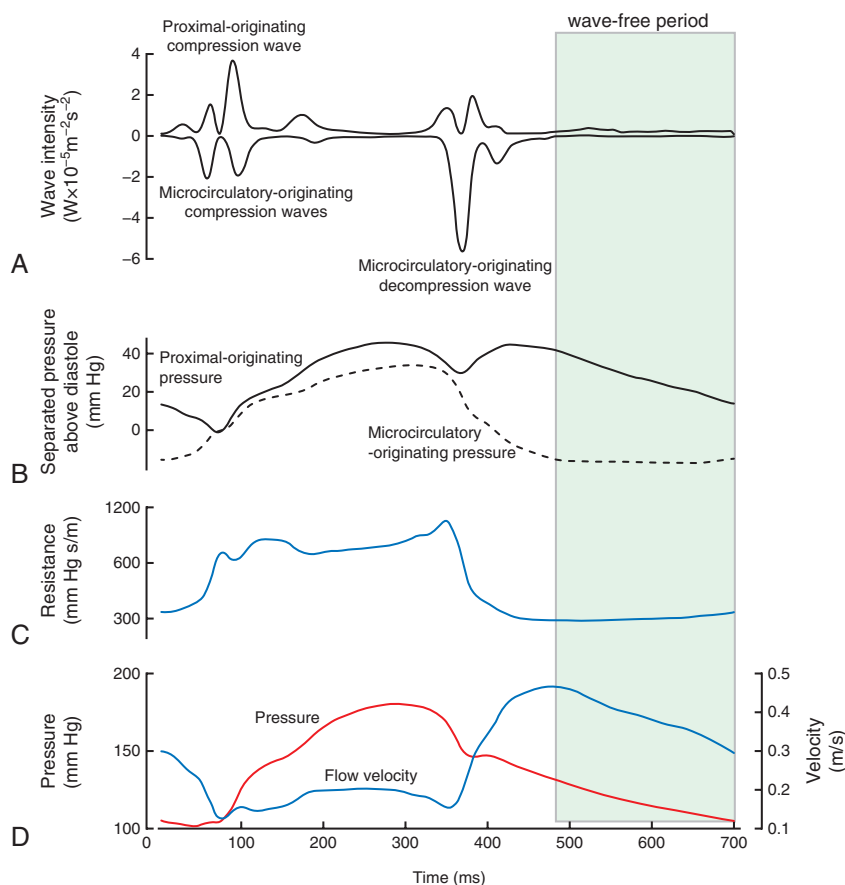


Fig. 14.18 The wave-free period during diastole is a phase when resistance is minimal (chart C), allowing for the determination of the instantaneous wave-free ratio. (From Sen S, Escaned J, Malik IS, et al. Development and validation of a new adenosine-independent index of stenosis severity from coronary wave-intensity analysis: results of the ADVISE (Adenosine Vasodilator Independent Stenosis Evaluation) study. *J Am Coll Cardiol.* 2012;59:1392–1402.)

“gray” zone. The negative predictive value is highest and correlates strongly with an FFR >0.80 if resting Pd/Pa is >0.96 and the positive predictive value is highest and correlates strongly with an FFR <0.80 if resting Pd/Pa is <0.87.⁷⁴ Thus values between 0.88 and 0.95 are indeterminate and should undergo a formal FFR assessment. This index is susceptible to wire drift, loading conditions, and hemodynamic variability compared to FFR or iFR; is less reproducible on serial measurements; and cannot be used to determine the significance of individual lesions in cases of serial coronary stenoses.

DIASTOLIC HYPEREMIA-FREE RATIO

The DFR is a proprietary index provided by Boston Scientific (Marlborough, Massachusetts). This is a resting index that uses the mean P_d/P_a during the period where the P_a is downsloping and less than the mean P_a . An electrocardiogram is not required for signal analysis, and the index is averaged over five cardiac cycles. A DFR <0.90 is considered significant and correlates strongly with iFR.⁷⁵

DIASTOLIC PRESSURE RATIO (DPR)

There are actually two technologies using the term “diastolic pressure ratio”: DPR and dPR. DPR is the ratio of the mean diastolic pressure distal to the stenosis to the mean diastolic aortic pressure. This value averages over the whole period of diastole, correlates with iFR, and a DPR <0.90 indicates a significant lesion.⁷⁶ Using software developed at Erasmus Medical Center, dPR is calculated during the flat period on the dP/dt curve of aortic pressure. At this point, resistance is constant and negligible. A dPR of <0.90 is considered significant and correlated well with iFR and FFR.^{77,78}

RESTING FULL CYCLE RATIO

Unlike diastolic-only indices, the RFR uses the lowest instantaneous P_d/P_a over the entire cardiac cycle. RFR requires proprietary software, and at least five consecutive cardiac cycles are needed to determine the RFR. The RFR value can be obtained or calculated offline, if needed, via waveform analysis. Using a cutoff value of <0.89 , RFR correlated well with FFR and iFR.⁷⁹

ANGIOGRAPHY-BASED INVASIVE INDICES

To avoid the risk of instrumentation of a coronary artery, angiography-based methodologies have been developed to determine the hemodynamic significance of stenoses. Quantitative flow ratio (QFR) is the angiography-based index with the largest evidence base. QFR simulates FFR and, as such, the same significance thresholds are used. Proprietary software is used to create three-dimensional coronary reconstructions by combining two angiographic projections at least 25 degrees apart. Flow equations based on the contrasted frame counts are then applied to the model. In clinical studies QFR correlates well with FFR⁸⁰ and avoids the risk, cost, and time associated with pressure wire placement into a coronary. However, QFR has limited utility in the assessment of significantly tortuous vessels, ostial lesions, or overlapping vessels. Numerous additional angiographic-based methods using proprietary software are being studied,⁷⁵ and it is possible that, as these angiographic-based methods evolve and improve, they will overshadow the invasive methods of determining lesion significance.

The various methods for determining coronary lesion significance are summarized in Table 14.1.

USE OF HEMODYNAMIC MEASUREMENTS DURING THE EVALUATION OF PERIPHERAL AND RENAL ARTERIAL DISEASE

Similar to coronary artery disease, angiography forms the basis of most decisions regarding revascularization in patients with peripheral arterial disease, and, similar to coronary artery disease, this approach is entirely adequate when the artery under investigation appears normal or severely stenosed. Not uncommonly, however, ambiguous lesions on the angiogram with moderate degrees of narrowing lead to difficult angiographic interpretation. In such cases, the measurement of a translesional pressure gradient provides useful information regarding the hemodynamic impact of the lesion.

A translesional pressure gradient can be measured at the time of angiography by one of several techniques. The simplest method samples pressure from the tip of a small-caliber (4-Fr) catheter, while the catheter is pulled back across the stenosis (Fig. 14.19). This method, however, is not ideal because the pullback technique is subject to beat-to-beat fluctuations in pressure from arrhythmia or the respiratory cycle. A better method involves the measurement of simultaneous pressures from two catheters positioned on either side of the stenosis. Although more accurate than the pullback technique, the use of even a small caliber catheter placed across the stenosis to measure a pressure gradient should be interpreted cautiously because the profile of the catheter across the lesion contributes to luminal obstruction, falsely elevating the gradient. Methods that avoid this problem include the positioning of two catheters on either side of the stenosis by obtaining an additional arterial access site (Fig. 14.20) or using a very small diameter (0.014-inch) wire outfitted with a pressure transducer near the catheter tip (i.e., a pressure wire). This arrangement is particularly useful for the assessment of renal artery lesions (Fig. 14.21). The waveform appearance of the pressure distal to a stenosis typically appears damped with a diminished peak systolic pressure and a delayed upstroke (Fig. 14.22B).

TABLE 14.1 Comparison of Different Methods to Assess the Significance of a Coronary Artery Lesion

INDEX	ISCHEMIC CUT-OFF VALUE	HYPER-EMIA?	PROPRIETARY SOFTWARE?	VALIDATED OUT-COMES?
FFR	<0.80	Yes	No	Yes
cFFR	<0.83	Yes	No	No
iFR	<0.90	No	Yes	Yes
Resting P_d/P_a	$<0.92-0.93$	No	No	No
DFR	<0.90	No	Yes	No
dPR	<0.90	No	Yes	No
RFR	<0.89	No	Yes	No
QFR	<0.80	No	Yes	No

cFFR, Contrast fractional flow reserve; dPR, diastolic pressure ratio; DFR, diastolic hyperemia-free ratio; FFR, fractional flow reserve; iFR, instantaneous wave-free ratio; RFR, resting full cycle ratio; P_a , pressure in the aorta; P_d , distal pressure; QFR, quantitative flow ratio.



Fig. 14.19 An example of obtaining a pullback pressure gradient by using a 4-Fr catheter placed distal to a subclavian stenosis (*left side*). When the catheter is withdrawn across the lesion, the systolic pressure is noted to rise by about 40 mm Hg (*arrow*).

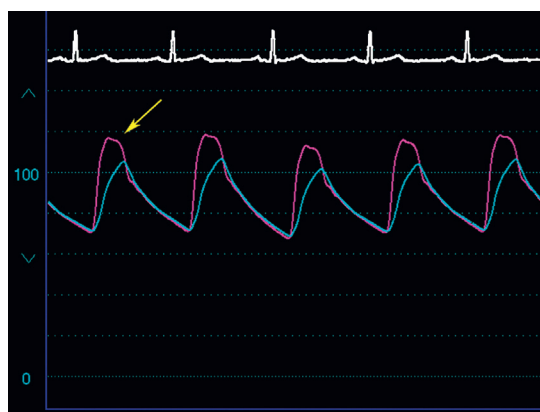


Fig. 14.20 These tracings were obtained in a patient with a right iliac stenosis. One catheter was positioned in the descending aorta via the left femoral artery to measure aortic pressure (*arrow*), while another catheter was placed below the stenosis in the common femoral artery. A 20-mm Hg systolic pressure gradient is noted. Observe also the damped appearance and delayed upstroke on the femoral artery pressure wave, consistent with a significant stenosis.

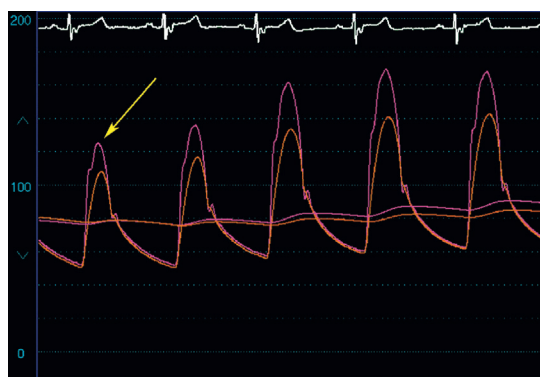


Fig. 14.21 These pressure tracings were obtained to assess the pressure gradient across a renal artery of moderate severity. Simultaneous pressure measurements were obtained from a catheter in the aorta (*arrow*) and from a 0.014-inch pressure wire. The gradient is 20–30 mm Hg. Note also the variability in the systolic pressure because of the respiratory cycle. For this reason, pullback pressures are suboptimal compared to simultaneous pressures.

Despite the widespread incorporation of translesional pressure gradients as an adjunct to angiography for decision-making, the actual *cutoff* value that constitutes a hemodynamically significant stenosis is not known and is still debated. Experts disagree regarding the choice of a systolic gradient, mean gradient, or hyperemic pressure gradient; many consider that a mean gradient of 10 mm Hg before or after vasodilators or a peak systolic gradient of 10 to 20 mm Hg should be considered a significant gradient.⁸¹

An early study correlating peripheral arterial angiography and hemodynamics found pressure gradients >20 mm Hg across most lesions narrowed by >75% by angiography, but not across lesions narrowed by <50%.⁸² Lesions that narrowed the lumen by 50% to 75% had a wide spectrum of pressure gradients, indicating that hemodynamic assessment is most helpful in this group. Measurement of pressure gradients helps not only with diagnosing a significant stenosis but also with determining the success of intervention (Fig. 14.22). In a study of balloon angioplasty of the iliac arteries, despite the appearance of a less-than-perfect angiographic result, the average peak systolic pressure gradient decreased from 44 to 7 mm Hg after successful balloon dilatation and resulted in clinical improvement.⁸³

In patients with PAD, the resting translesional pressure gradient may not be indicative of the hemodynamic consequences of the lesion during exercise when blood flow is increased. This should be considered when luminal narrowing is moderate by angiography and the pressure gradient at rest is <20 mm Hg, but the patient has convincing symptoms with exertion. The use of a vasodilator such as nitroglycerin may prove useful, as exemplified in the published case of a patient with moderate narrowing of the common iliac artery on angiography and a resting gradient increasing from 6 to 22 mm Hg with intraarterial nitroglycerin.⁸⁴

Pressure gradient determination has been most commonly applied to the evaluation of renal artery lesions (Fig. 14.23). Early studies used the bulky 4-Fr catheters to measure the translesional pressure gradients in renal arteries, and this is problematic in the smaller caliber renal arteries because the catheter contributes to obstruction. One study that used both 4-Fr catheters and the unobtrusive 0.014-inch pressure wire to measure pressure gradients in the same arterial lesions found an average systolic pressure gradient of 40 ± 46 mm Hg by pressure wire compared with 71 ± 52 mm Hg by 4-Fr catheter. Thus the 4-Fr catheter clearly overestimated the systolic gradient, suggesting that the

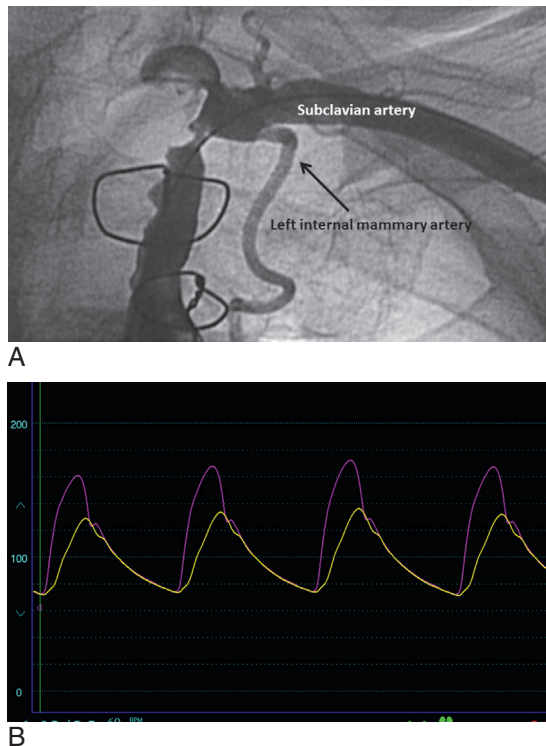


Fig. 14.22 (A) A 78-year-old male with prior bypass surgery with a left internal mammary artery graft to the left anterior descending artery developed progressive angina and is found to have a severe stenosis of the subclavian artery proximal to the left internal mammary artery (arrow). (B) There is a significant pressure gradient across the stenosis.

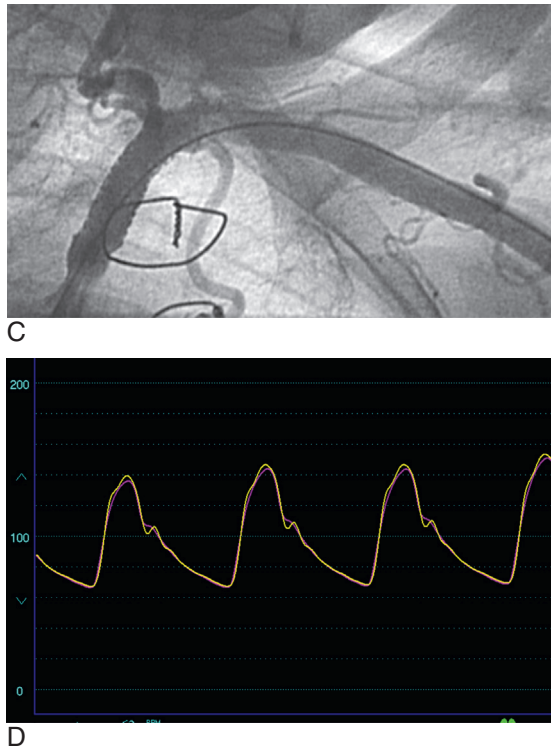


Fig. 14.22, cont'd. (C) The lesion is stented successfully with an excellent angiographic result. (D) There is no residual gradient noted after stenting.



Fig. 14.23 These tracings demonstrate the use of a pressure guidewire to assess a renal artery lesion and the result of an intervention. (A) Simultaneous pressure measurements were obtained from a catheter in the aorta and from a 0.014-inch pressure wire and demonstrated a 30-mm Hg peak systolic gradient.

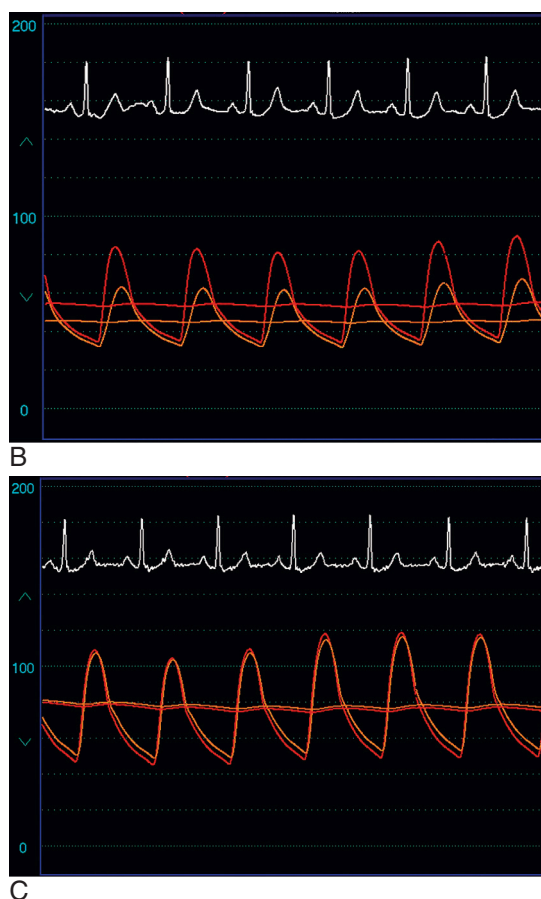


Fig. 14.23, cont'd (B) Intrarenal nitroglycerin was administered and lowered systolic pressure but did not change the peak systolic gradient. (C) Following treatment of the lesion with a stent, the gradient was eliminated.

pressure wire is a superior technique for measuring transluminal pressure gradients.⁸⁵ If the pressure gradient in the renal artery is <20 mm Hg, vasodilator agents (dopamine, nitroglycerine, or papaverine) can be used to augment the pressure gradient. While adenosine administration to the coronaries causes vasodilation, it cannot be used in the renal arteries for this purpose since it causes renal arterial vasoconstriction. Similar to other peripheral vascular beds, the translesional gradient representing a “hemodynamically significant” renal artery lesion is not clear. An elegant study by De Bruyne et al.⁸⁶ analyzed the hemodynamics of varying degrees of renal artery obstruction created by partial balloon inflation in the renal artery following a renal artery stenting procedure. With a pressure wire, the ratio of pressure distal to the stenosis created by the partially inflated balloon (P_d) to pressure in the aorta (P_a) was compared to renin levels sampled from the renal vein. Renin levels rose above normal at a P_d/P_a of <0.9 and correlated with the degree of obstruction. Another study comparing resting and hyperemic pressure gradients induced by intraarterial papaverine found that a hyperemic systolic gradient of >21 mm Hg or a renal FFR of <0.90 was associated with renin production, indicating hemodynamic significance.⁸⁷ Another study by Drieghe et al. found that renal artery stenosis on renal angiography of $>50\%$ and minimal lumen diameter of <2 mm was associated with a P_d/P_a of <0.90 .⁸⁸

In terms of outcome data following stenting, both a papaverine-induced hyperemic systolic gradient of >21 mm Hg⁸⁹ and a dopamine-induced mean gradient of >20 mm Hg⁹⁰ predicted the improvement in blood pressure after renal stenting. Disappointingly the Cardiovascular Outcomes in Renal Atherosclerotic Lesions (CORAL) trial (a study that randomized patients to renal stenting versus optimal medical therapy for renal artery stenosis) did not find a benefit with renal artery stenting, even in patients with high transluminal pressure gradients.⁹¹ In aggregate, these data lead one to question the role of this technology in determining the need for renal artery intervention.

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